[INCH-POUND]
A-A-59400
26 October 1999
SUPERSEDING
MIL-T-43560A
1 November 1972

COMMERCIAL ITEM DESCRIPTION

TESTER, SPRING RESILIENCY (PUSH/PULL FORCE GAUGE)

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-T-43560A for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for manually operated spring resiliency testers (commercially called push/pull force gauges). These testers will be used to measure and provide a numerical indication of the tension and compression strengths (resiliency) of springs and spring-loaded devices.

2. CLASSIFICATION. The testers supplied are classified according to the types, classes, and sizes listed below and in table I. The type, class, and size of the testers to be supplied will be specified in the acquisition order (see 7.5(b)). Tester sizes other than those listed in table I will be specified in the acquisition order (see 7.5(c)).

Type I	Linear indicator
Type II	Dial indicator
Type III	Digital electronic indicator
Class A	Light duty
Class B	Heavy duty
Size	Various (see table I)

3. SALIENT CHARACTERISTICS

3.1 General requirements. The tester shall be direct reading and capable of both tension and compression measurements. The Type II (dial) and Type III (digital electronic) shall capable of being hand held or used on a test stand.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610

AMSC N/A FSC 6635

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TABLE I. Spring resiliency testers.

Type	Class	Size	Capacity Range	Scale Graduation
I	A	1	0-50 g	2 g
		2	0-100 g	4 g
		3	0-8 oz	0.25 oz
		4	0-2 lb	0.50 oz
	B	1	0-20 lb	4 oz
		2	0-80 lb	1 lb
II	A	1	0-500 g	5 g
		2	0-1 kg	10 g
		3	0-10 kg	100 g
		4	0-16 oz	0.125 oz
		5	0-5 lb	0.05 lb
		6	0-10 lb	0.10 lb
		7	0-20 lb	0.20 lb
		8	0-30 lb	0.20 or 0.25 lb
	B	1	0-50 lb	0.50 lb
		2	0-100 lb	1 lb
		3	0-150 lb	1 lb
		4	0-200 lb	2 lb
		5	0-250 lb	2.5 lb
III	A	1	0-5 oz 0-250 g	0.001 oz 0.1 g
		2	0-1 lb 0-500 g	0.001 lb 0.1 g
		3	0-2 lb 0-1 kg	0.001 lb 0.001 kg
		4	0-10 lb 0-5 kg	0.005 lb 0.005 kg
		5	0-20 lb 0-10 kg	0.01 lb 0.01 kg
	B	1	0-50 lb 0-25 kg	0.01 lb 0.01 kg
		2	0-100 lb 0-50 kg	0.1 lb 0.01 kg
		3	0-250 lb 0-110 kg	0.1 lb 0.1 kg
		4	0-500 lb 0-220 kg	0.1 lb 0.1 kg

3.2 Indicator. The spring force indicator shall move along a linear or circular (dial) graduated scale or shall provide a digital electronic readout. Type I (linear) and Type II (dial) testers may

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include dual scales for ounce/pound and gram/kilogram measurements. Type III (digital electronic) testers shall provide both ounce/pound and gram/kilogram readouts.

3.2.1 Peak reading. The tester shall include an indicator-locking device which shall maintain the maximum indicator reading after the tested force is removed.

3.2.2 Accuracy. The Type I testers shall be accurate within one scale graduation for loads axially and at an angle up to two degrees from the shaft of the axis. The Type II and Type III testers shall be accurate within one percent of full scale capacity for loads axially and at an angle up to two degrees from the shaft of the axis.

3.3 Power requirements. Electrical power supplies for Type III testers shall be specified in the acquisition order (see 7.5(d)).

3.4 Attachments. The testers shall be furnished with the following stainless steel attachments:

a. Flat head
b. Notched head
c. Cone head
d. Chisel head
e. Hook
f. Extension rod, 6-inch length

3.5 Accessories. If specified in the acquisition order (see 7.5(e)), accessories such as handles, cradles, grippers, and test stand adapters shall be provided.

3.6 Overload protection. The tester shall be equipped with an overload protector to prevent damage to the tester during force application.

3.7 Zero adjustment. The tester shall have a means of zeroing the indicator.

3.8 Calibration certificate. A calibration certificate shall be supplied with each tester.

3.9 Handles. The Type II tester shall be supplied with removable handles.

3.10 Shaft ends. The Type II tester shall have threaded shaft ends with 10-32 threads (Class A) and 5/16-18 threads (Class B).

3.11 Insulating link. If specified in the acquisition order (see 7.5(f)), an insulating link shall be provided for protection from shock when testing live electrical contacts.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The testers provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product as that which the producer offers for sale in the commercial market. The government reserves the right to require proof of such conformance.

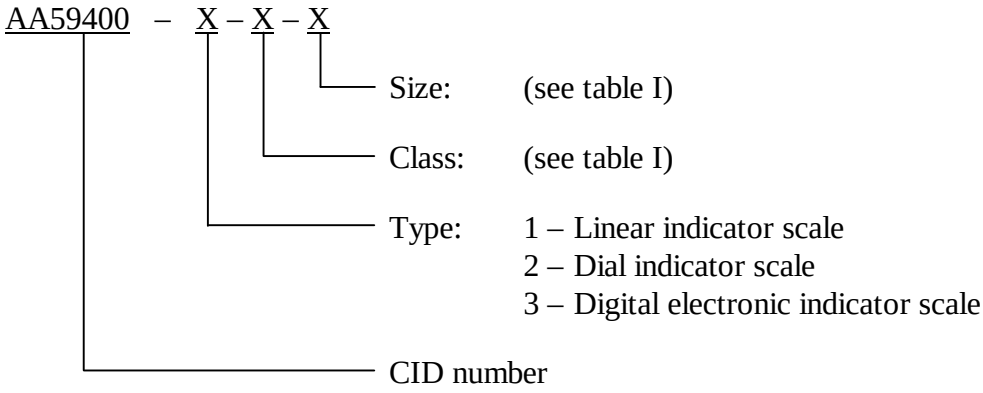
5.2 Market acceptability. The tester offered must have been on the commercial market for at least 2 years. Significant quantities must have been sold to commercial or military customers.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the testers shall be preserved, packed, and marked as specified in the acquisition order (see 7.5(g)).

7. NOTES

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Defense Automated Printing Service, Building 4D, 700 Robbins Avenue, Philadelphia, PA, 19111-5094.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply spring resiliency testers that meet the salient characteristic requirements of this CID. Competition is not limited to the listed firms.

Wagner Instruments
PO Box 1217
Greenwich, CT 06836

AMETEK, Test and Calibration Instruments
8600 Somerset Drive
Largo, FL 33773

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7.4 Cross-reference. The following is a cross-reference between the classification types in MIL-T-43560A and this CID:

TABLE II. Type cross-reference.

MIL-T-43560A	A-A-59400 Type	Description
Type I, Class 1, Style A Type I, Class 1, Style B	Type I, Class A Type I, Class B	Linear indicator, light duty Linear indicator, heavy duty
Type I, Class 2, Style A Type I, Class 2, Style B	Type II, Class A Type II, Class B	Dial indicator, light duty Dial indicator, heavy duty
Type II	Not covered	Tension (pull) testers
Not covered	Type III, Class A Type III, Class B	Digital electronic indicator, light duty Digital electronic indicator, heavy duty

7.5 Ordering data. Acquisition documents must specify the following requirements:

a. Title, number, and date of this document
b. Classification (type, class, size) (see table I)
c. Capacity range and scale graduation (specify if different than table I)
d. Power requirements (see 3.3)
e. Accessories, if required (see 3.5)
f. Insulating link, if required (see 3.11)
g. Packaging requirements (see 6.1)

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - GL
Navy - AS

Reviewer:
Army - CR4

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS

(Project 6635-0185)

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